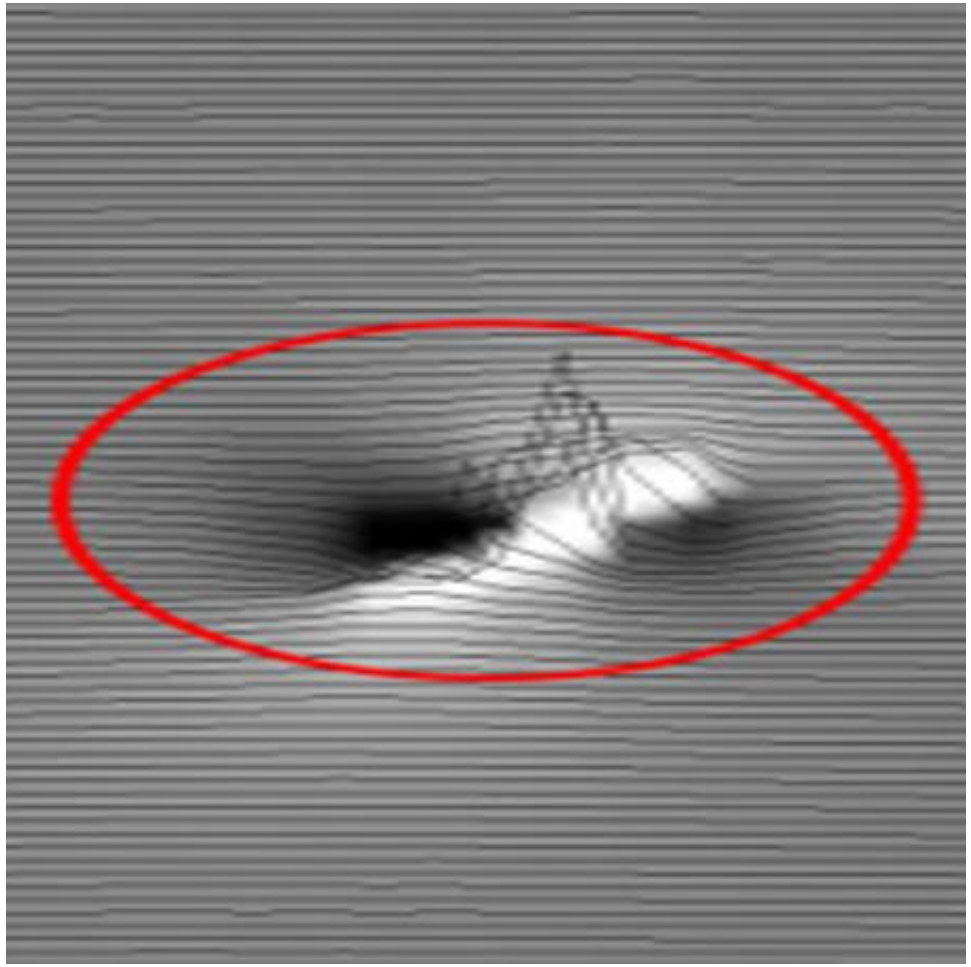
DEEP LEARNING-BASED DEFECT DETECTION AND LOCALIZATION IN STEEL SURFACE IMAGES

RUSTY JOHNSTON – ECEN 5743 FINAL PROJECT – SPRING ‘26



MOTIVATION

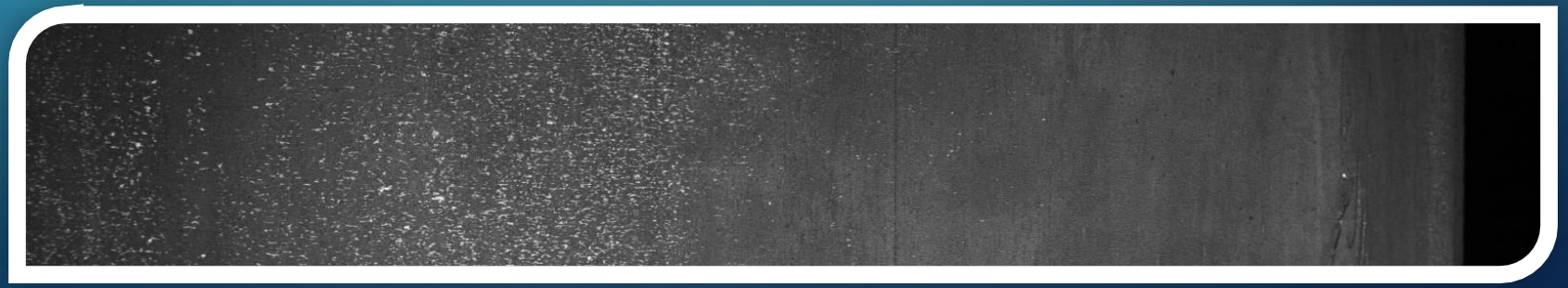
- Magnetic Flux Leakage (MFL) used for pipeline defect detection
- Produces grayscale “field maps” of material anomalies
- Requires:
  - Defect detection
  - Defect localization

PROJECT ANALOGIES

- Steel surface images ≈ MFL signal maps
- Defects appear as spatial patterns in grayscale data
- Similar ML task:
  - Identify presence of defects
  - Locate defect regions

DATASET: SEVERSTAL STEEL DEFECT DETECTION

- Kaggle competition dataset
- High-resolution grayscale steel surface images
- Pixel-level defect annotations (RLE encoded masks)
- 4 defect classes + non-defective images



APPROACH OVERVIEW

- Step 1: Binary classification (CNN)
- Step 2: Multiclass segmentation (U-Net, 4 classes)
- Step 3: Binary segmentation (U-Net, defect vs background)
- Framework:
  - PyTorch
  - GPU: NVIDIA A10G

CLASSIFICATION MODEL

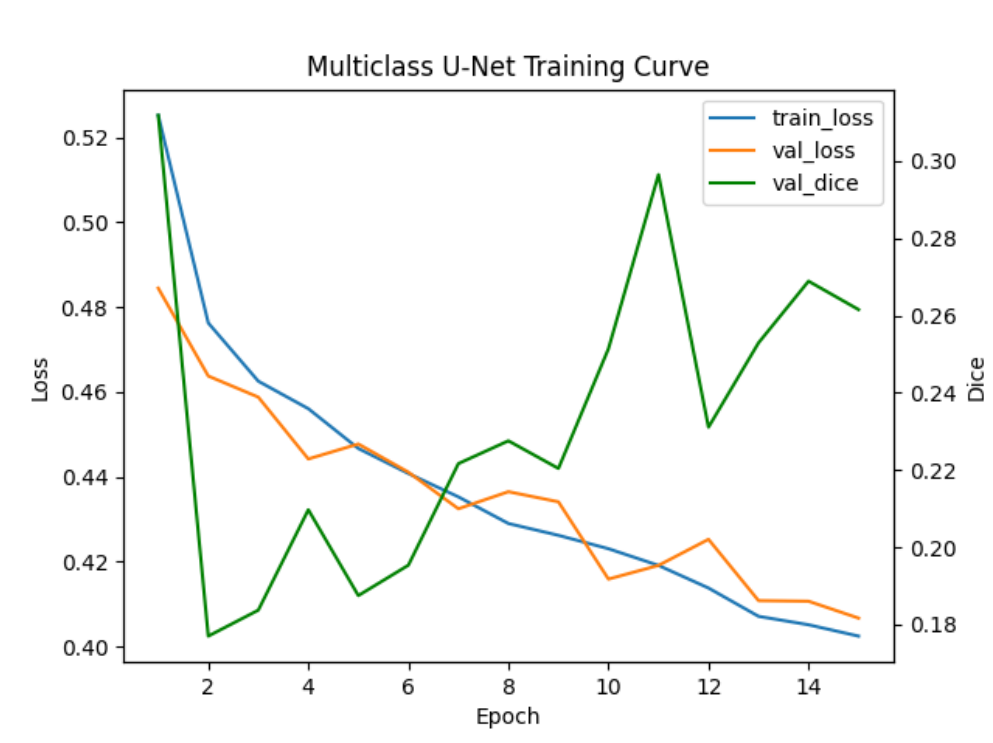
- Input: 128 x 128 grayscale images
- CNN architecture
- Output: defect / no defect
- Metric: F1 score

- Results:
  - F1= 0.82
  - Accuracy = 0.80

MULTICLASS SEGMENTATION (U-NET)

- Output: 4-channel mask (one per defect class)
- Loss: BCEWithLogitsLoss
- Metric: Dice Coefficient
- Issues:
  - Severe class imbalance
  - Sparse defect regions
- Results:
  - Dice = 0.32

MULTICLASS TRAINING BEHAVIOR

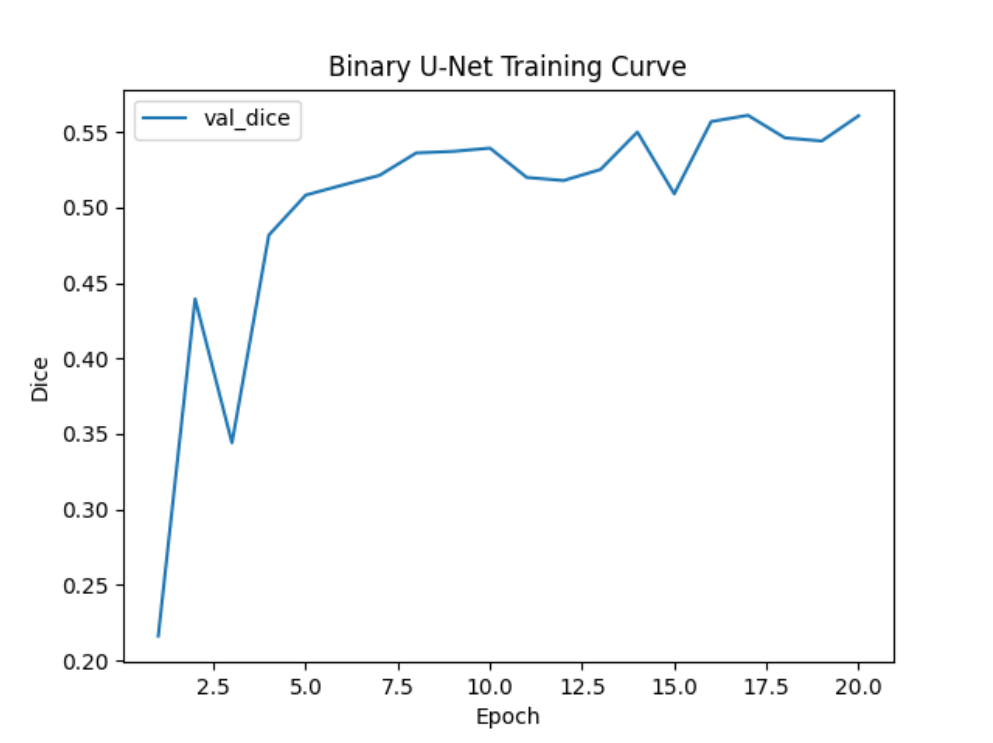


- Loss decreases steadily
- Dice climbs some but fluctuates and remains low
- Indication of difficulty learning class-specific masks

BINARY SEGMENTATION (U-NET)

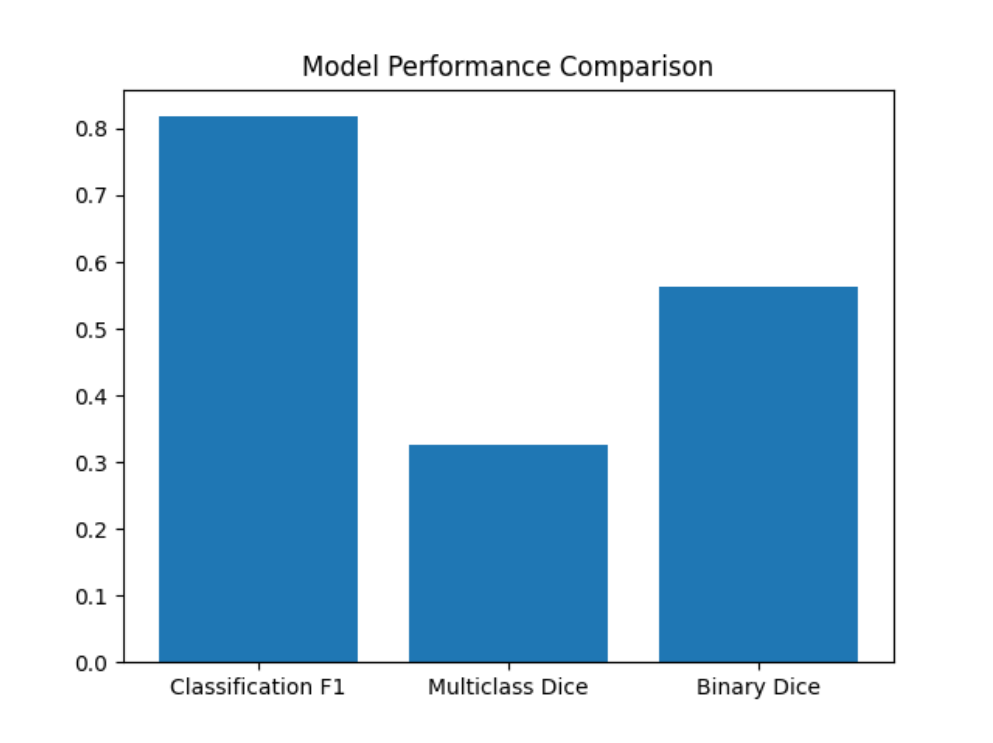
- Output: 1-channel mask (defect vs background)
- Loss: BCEWithLogitsLoss
- Metric: Dice Coefficient
- Advantages:
  - Reduced class imbalance
  - Stronger learning signal from combined defect regions
  - Better alignment with detection objective
- Results:
  - Dice = 0.56

BINARY TRAINING BEHAVIOR



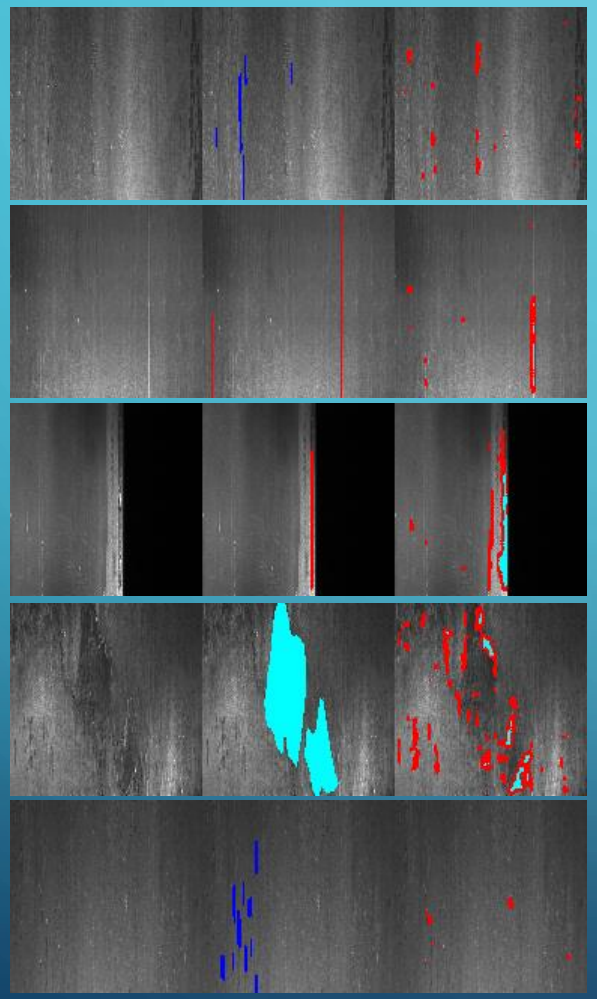
- Rapid early improvement
- Stable convergence around 0.55
- Much better learning signal

RESULTS COMPARISON

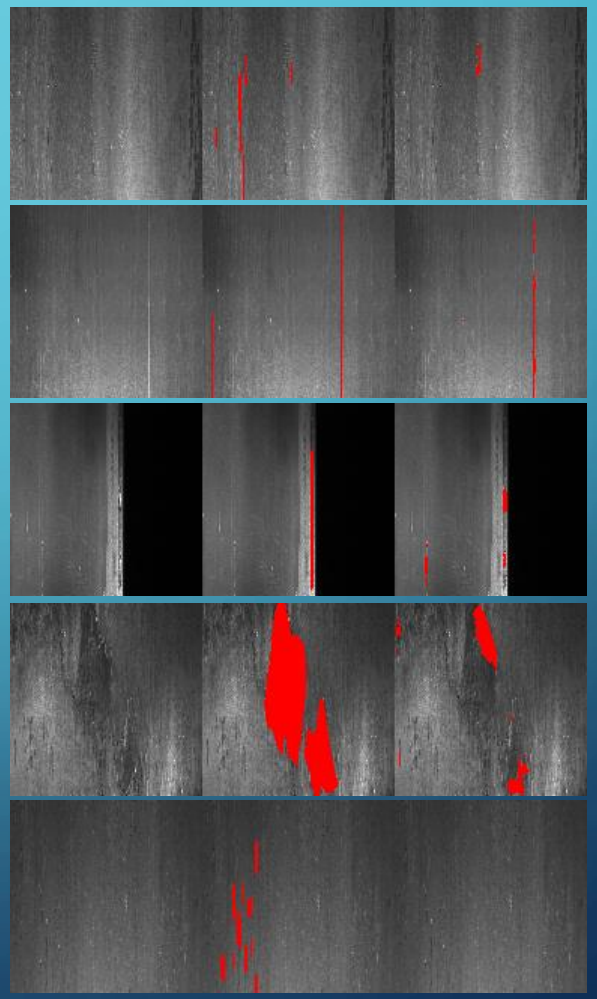


- Classification using CNN performs well with minimal adjustments
- Multiclass segmentation struggles
- Binary segmentation significantly improves target metric performance
- Important to distinguish target metric performance vs. problem solution

MULTICLASS VS BINARY EXAMPLES



Multiclass



Binary

DISCUSSION: TRADE OFFS IN SEGMENTATION

MULTICLASS

- Advantages
  - Can distinguish classes
- Limits
  - Class imbalance hurts performance
  - Sparse and inconsistent labels
  - Weak class-specific learning goal

BINARY

- Advantages
  - Stronger learning signal
  - Improved detection and spatial coherence
  - Higher metric performance
- Limits
  - Cannot distinguish defect types

TAKEAWAY

- Higher dice does not imply a better overall solution
- Model formulation must match end application

CONCLUSION

- Two successful examples of detection and localization pipeline
- Key metric improvement
- Strong baseline for improvement for future project
- Future work for improvement:
  - Data augmentation
  - High resolution inputs
  - Pretrained encoders
  - Multiclass strategies with improved class balancing